

Tasks Day 01

Sprint 3 Goal:

"Diagnose and eliminate N+1 query bottlenecks to reduce database round-trips, implement Redis cache-aside pattern for high-read catalog endpoints, and apply composite indexes to optimize execution plans, providing measured before/after performance evidence."

مهام ال Backlog لهذا الأسبوع:

- [Day 1] Enable EF Core logging and diagnose a genuine N+1 problem.
- [Day 2] Fix N+1 issues using Eager Loading (Include) or Projection (Select).
- [Day 3] Integrate Redis caching for the Patients catalog.
- [Day 4] Add composite database indexes for frequent queries.

[Performance Issue] N+1 Query Problem Identification & Documentation

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Sensitive data logging is enabled. Log entries and exception messages may include sensitive application data; this mode should only be enabled during development.
warn: 8/9/2026 20:18:56.148 CoreEventId.SensitiveDataLoggingEnabledWarning[10400] (Microsoft.EntityFrameworkCore.Infrastructure)
Sensitive data logging is enabled. Log entries and exception messages may include sensitive application data; this mode should only be enabled during development.

warn: Microsoft.EntityFrameworkCore.Query[10102]
The query uses a row limiting operator ('Skip'/'Take') without an 'OrderBy' operator. This may lead to unpredictable results. If the 'Distinct' operator is used after 'OrderBy', then make sure to use the 'OrderBy' operator after 'Distinct' as the ordering would otherwise get erased.
warn: 8/9/2026 20:24:37.237 CoreEventId.RowLimitingOperationWithoutOrderByWarning[10102] (Microsoft.EntityFrameworkCore.Query)
The query uses a row limiting operator ('Skip'/'Take') without an 'OrderBy' operator. This may lead to unpredictable results. If the 'Distinct' operator is used after 'OrderBy', then make sure to use the 'OrderBy' operator after 'Distinct' as the ordering would otherwise get erased.
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (144ms) [Parameters=[@__p_0='10'], CommandType='Text', CommandTimeout='30']
SELECT TOP(@__p_0) [p].[Id], [p].[ContactNumber], [p].[DateOfBirth], [p].[FirstName], [p].[Gender], [p].[LastName], [p].[UserId]
FROM [Patients] AS [p]
info: 8/9/2026 20:24:38.455 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
Executed DbCommand (144ms) [Parameters=[@__p_0='10'], CommandType='Text', CommandTimeout='30']
SELECT TOP(@__p_0) [p].[Id], [p].[ContactNumber], [p].[DateOfBirth], [p].[FirstName], [p].[Gender], [p].[LastName], [p].[UserId]
FROM [Patients] AS [p]
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (11ms) [Parameters=[@__patient_id_0='1'], CommandType='Text', CommandTimeout='30']
SELECT [v].[Id], [v].[DiastolicBP], [v].[HeartRate], [v].[OxygenSaturation], [v].[PatientId], [v].[RecordedAt], [v].[SystolicBP]
FROM [VitalSigns] AS [v]
WHERE [v].[PatientId] = @__patient_id_0
info: 8/9/2026 20:24:38.626 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
Executed DbCommand (11ms) [Parameters=[@__patient_id_0='1'], CommandType='Text', CommandTimeout='30']
SELECT [v].[Id], [v].[DiastolicBP], [v].[HeartRate], [v].[OxygenSaturation], [v].[PatientId], [v].[RecordedAt], [v].[SystolicBP]
FROM [VitalSigns] AS [v]
WHERE [v].[PatientId] = @__patient_id_0
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (1ms) [Parameters=[@__patient_id_0='2'], CommandType='Text', CommandTimeout='30']
SELECT [v].[Id], [v].[DiastolicBP], [v].[HeartRate], [v].[OxygenSaturation], [v].[PatientId], [v].[RecordedAt], [v].[SystolicBP]
FROM [VitalSigns] AS [v]
WHERE [v].[PatientId] = @__patient_id_0
info: 8/9/2026 20:24:38.638 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
Executed DbCommand (1ms) [Parameters=[@__patient_id_0='2'], CommandType='Text', CommandTimeout='30']
SELECT [v].[Id], [v].[DiastolicBP], [v].[HeartRate], [v].[OxygenSaturation], [v].[PatientId], [v].[RecordedAt], [v].[SystolicBP]
FROM [VitalSigns] AS [v]
WHERE [v].[PatientId] = @__patient_id_0
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (1ms) [Parameters=[@__patient_id_0='3'], CommandType='Text', CommandTimeout='30']
SELECT [v].[Id], [v].[DiastolicBP], [v].[HeartRate], [v].[OxygenSaturation], [v].[PatientId], [v].[RecordedAt], [v].[SystolicBP]
FROM [VitalSigns] AS [v]
WHERE [v].[PatientId] = @__patient_id_0
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WHERE [v].[PatientId] = @__patient_Id_0
nfo: 8/9/2026 20:24:38.668 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
Executed DbCommand (1ms) [Parameters=[@__patient_Id_0='5'], CommandType='Text', CommandTimeout='30']
SELECT [v].[Id], [v].[DiastolicBP], [v].[HeartRate], [v].[OxygenSaturation], [v].[PatientId], [v].[RecordedAt], [v].[SystolicBP]
FROM [VitalSigns] AS [v]
WHERE [v].[PatientId] = @__patient_Id_0
nfo: Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (1ms) [Parameters=[@__patient_Id_0='6'], CommandType='Text', CommandTimeout='30']
SELECT [v].[Id], [v].[DiastolicBP], [v].[HeartRate], [v].[OxygenSaturation], [v].[PatientId], [v].[RecordedAt], [v].[SystolicBP]
FROM [VitalSigns] AS [v]
WHERE [v].[PatientId] = @__patient_Id_0
nfo: 8/9/2026 20:24:38.723 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
Executed DbCommand (1ms) [Parameters=[@__patient_Id_0='6'], CommandType='Text', CommandTimeout='30']
SELECT [v].[Id], [v].[DiastolicBP], [v].[HeartRate], [v].[OxygenSaturation], [v].[PatientId], [v].[RecordedAt], [v].[SystolicBP]
FROM [VitalSigns] AS [v]
WHERE [v].[PatientId] = @__patient_Id_0
nfo: Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (1ms) [Parameters=[@__patient_Id_0='7'], CommandType='Text', CommandTimeout='30']
SELECT [v].[Id], [v].[DiastolicBP], [v].[HeartRate], [v].[OxygenSaturation], [v].[PatientId], [v].[RecordedAt], [v].[SystolicBP]
FROM [VitalSigns] AS [v]
WHERE [v].[PatientId] = @__patient_Id_0
nfo: 8/9/2026 20:24:38.731 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
Executed DbCommand (1ms) [Parameters=[@__patient_Id_0='7'], CommandType='Text', CommandTimeout='30']
SELECT [v].[Id], [v].[DiastolicBP], [v].[HeartRate], [v].[OxygenSaturation], [v].[PatientId], [v].[RecordedAt], [v].[SystolicBP]
FROM [VitalSigns] AS [v]
WHERE [v].[PatientId] = @__patient_Id_0
nfo: Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (1ms) [Parameters=[@__patient_Id_0='8'], CommandType='Text', CommandTimeout='30']
SELECT [v].[Id], [v].[DiastolicBP], [v].[HeartRate], [v].[OxygenSaturation], [v].[PatientId], [v].[RecordedAt], [v].[SystolicBP]
FROM [VitalSigns] AS [v]
WHERE [v].[PatientId] = @__patient_Id_0
nfo: 8/9/2026 20:24:38.739 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
Executed DbCommand (1ms) [Parameters=[@__patient_Id_0='8'], CommandType='Text', CommandTimeout='30']
SELECT [v].[Id], [v].[DiastolicBP], [v].[HeartRate], [v].[OxygenSaturation], [v].[PatientId], [v].[RecordedAt], [v].[SystolicBP]
FROM [VitalSigns] AS [v]
WHERE [v].[PatientId] = @__patient_Id_0
nfo: Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (1ms) [Parameters=[@__patient_Id_0='9'], CommandType='Text', CommandTimeout='30']
SELECT [v].[Id], [v].[DiastolicBP], [v].[HeartRate], [v].[OxygenSaturation], [v].[PatientId], [v].[RecordedAt], [v].[SystolicBP]

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